

# Jiechao Gao

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| CONTACT INFORMATION     |  +1 929-213-7363<br> jiechao@stanford.edu<br> linkedin.com/in/jiechaogao<br> Scholar<br> Homepage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Y2E2<br>Center for Sustainable Development & Global Competitiveness<br>Stanford University<br><a href="#">Stanford Profile</a><br>Stanford, CA |
| STATUS                  | <b>On the 2026–27 academic job market. U.S. Permanent Resident, no visa sponsorship required.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                |
| RESEARCH INTERESTS      | <p>I build translational AI systems: systems that hold up in hospitals, buildings and power grids, financial markets, and the platforms people use every day. Three things have to be true before that is possible, and each is one of my research lines:</p> <ul style="list-style-type: none"><li>• <b>Trustworthy Generation &amp; Agents:</b> generation and agents whose output can be checked against the structure of its target; structural priors in generators, self-correcting agents, evaluators and judges (NeurIPS'26 ×4, ICML'26 ×2, ICLR'26, AACL'26 ×2, WWW'26, ACL'26, Findings of ACL'26)</li><li>• <b>Alignment at Large Scale:</b> representation alignment and calibration for learning heterogeneous data in one model, validated in production A/B tests (NeurIPS'25, SIGIR'26, KDD'25 ×2, ICML'25, ACM MM'25 ×2)</li><li>• <b>Privacy-Preserving &amp; Efficient Learning:</b> federated learning and efficient inference where data cannot leave the device and compute is scarce; differential privacy, on-device personalization, economic viability of LLM deployment (NPJ AI'26, ICCPS'25, SenSys, IEEE TSC'20, IPDPS'20, EMNLP'25)</li><li>• <b>Translational AI Systems &amp; Open Source:</b> where the three are put to the test: clinical reports and sensing in hospitals, buildings and grids, financial markets and open source (FinRL ecosystem), generated software, and recommendation platforms serving hundreds of millions of people (AAAI'26, EMNLP'25 Best Paper Nom., CHIL'23, BuildSys'24, NeurIPS'22 D&amp;B)</li></ul> |                                                                                                                                                |
| PROFESSIONAL EXPERIENCE | <b>Stanford University</b><br>Postdoc Research Associate<br>Jul. 2025 - now <ul style="list-style-type: none"><li>• Research on trustworthy generative models and resource-efficient LLM systems, with applications in engineering and sustainability.</li><li>• <b>Advisors:</b> Prof. Jie Wang and Prof. Michael Lepech</li></ul><br><b>Columbia University</b><br>Associate Research Scientist<br>Aug. 2024 - June 2025 <ul style="list-style-type: none"><li>• Federated learning, reinforcement learning, and trustworthy LLM agents for financial applications; core contributor and contest organizer of the FinRL open-source ecosystem.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                |
| INDUSTRY EXPERIENCE     | <b>Frees Fund (VC), Beijing</b><br>AI Investment Intern<br>May 2018 - Aug. 2018                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                |
| EDUCATION               | <b>Ph.D. in Computer Science</b><br><i>University of Virginia</i><br>Department of Computer Science <ul style="list-style-type: none"><li>• <b>Advisor:</b> Prof. Bradford Campbell</li></ul><br><b>Master of Science in Electrical Engineering</b><br><i>Columbia University</i><br>Department of Electrical Engineering<br>Sep. 2016 - Feb. 2018<br>New York, NY<br><br><b>B.S. in Applied Physics &amp; B.Eng. in Financial Engineering</b><br><i>Jilin University</i><br>Department of Physics & Department of Economics<br>Sep. 2012 - Jul. 2016<br>Changchun, Jilin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                |

## AWARDS

|                                                             |             |
|-------------------------------------------------------------|-------------|
| Google HE Faculty AI Fellowship                             | 2026        |
| EMNLP Industry Track Best Paper Nomination                  | 2025        |
| World's Top 2% Scientists (Stanford/Elsevier)               | 2024 & 2025 |
| ACM BuildSys Best Paper Candidate                           | 2024        |
| ICPP NSF Travel Award                                       | 2023        |
| UVA Link Lab Flash Talk Awardee                             | Feb. 2022   |
| Excellent Graduation Thesis (5/242), JLU Physics Department | Jun. 2016   |
| MCM Outstanding Winner Award                                | Apr. 2015   |
| National Scholarship (Top 1%)                               | 2013 & 2014 |
| Manulife Financial Center Outstanding Intern Award          | 2014        |

## PUBLICATIONS



**4,758 Citations · H-index 29 · i10-index 56** (Google Scholar, as of Sep. 2026)

**110 peer-reviewed publications**, with **70 as first / co-first / corresponding author**. Venues include **NeurIPS ×6, ICML ×3, ICLR, AAAI ×2, KDD ×3, WWW, SIGIR, ACM MM ×3, CVPR, the ACL family ×12 (ACL / EMNLP / EACL, incl. Industry & Findings), NPJ Artificial Intelligence, ICCPS, CHIL, IPDPS, and IEEE TSC**; with 1 Best Paper Nomination (EMNLP Industry '25), 1 Best Paper Candidate (BuildSys '24), and 2 Orals (ACM MM '25 top 4%; CHIL '23 13.3%).

\* equal contribution    † corresponding author

### Peer-reviewed Conference Papers (with proceedings)

|             |                                                                                                                                                                                                                                                                                                 |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NeurIPS '26 | 66. Haoran Ma, Linxiao Li, Chenyue Wang, Haochen Sui, <b>Jiechao Gao</b> <sup>†</sup><br>Phase-wise MLLM Tuning for Multi-framework WebUI Code Generation<br><i>Conference on Neural Information Processing Systems</i> , 2026                                                                  |
| NeurIPS '26 | 65. Linxiao Li, Haoran Ma, Chenyue Wang, Jiaye Lin, Haochen Sui, <b>Jiechao Gao</b> <sup>†</sup><br>Toward Executable Multi-framework Front-end Code Generation with Self-Correction<br><i>Conference on Neural Information Processing Systems</i> , 2026                                       |
| NeurIPS '26 | 64. Haoran Ma*, <b>Jiechao Gao</b> *, Shisong Tang, Bing Han, Jifeng Hu, Hechang Chen<br>Render Structure Uncertainty for HTML Repair in MLLM-based UI-to-Code Generation<br><i>Conference on Neural Information Processing Systems</i> , 2026                                                  |
| NeurIPS '26 | 63. <b>Jiechao Gao</b> , Chang Liu, Yuandong Pan, Ying Liu, Michael Lepech<br>Anchoring LLM-based Chest X-ray Report Generation via Diffusion Language Planning<br><i>Conference on Neural Information Processing Systems</i> , 2026                                                            |
| ICML '26    | 62. Zixiong Yang*, Linxiao Li*, Jiaye Lin, Binrui Wu, Xiaoyu Kang, <b>Jiechao Gao</b> <sup>†</sup><br>Deterministic Component Mining for Multi-framework UI2Code Generation<br><i>International Conference on Machine Learning</i> , 2026                                                       |
| ICML '26    | 61. Jie Wu*, Haoran Ma*, Shisong Tang, Yulin Xu, Xiaoyu Kang, <b>Jiechao Gao</b> <sup>†</sup><br>MulFCoder: Framework-conditioned Multi-agent for MLLM-based Multi-framework Front-end Code Generation<br><i>International Conference on Machine Learning</i> , 2026                            |
| ICLR '26    | 60. Siyuan Chen*, Hang Ding*, Kangxiao Yu, <b>Jiechao Gao</b> <sup>†</sup><br>GRO-RAG: Gradient-aware Re-rank Optimization for Multi-source Retrieval-Augmented Generation<br><i>International Conference on Learning Representations</i> , 2026                                                |
| AAAI '26    | 59. <b>Jiechao Gao</b> <sup>†</sup> , Chang Liu, Yuangang Li<br>S2D-ALIGN: Shallow-to-Deep Auxiliary Learning for Anatomically-Grounded Radiology Report Generation<br><i>AAAI Conference on Artificial Intelligence</i> , 2026                                                                 |
| AAAI '26    | 58. Yuangang Li*, Yiqing Shen*, Yi Nian, <b>Jiechao Gao</b> , Ziyi Wang, Chenxiao Yu, Shawn Li, Jie Wang, Xiyang Hu <sup>†</sup> , Yue Zhao <sup>†</sup><br>Mitigating hallucinations in large language models via causal reasoning<br><i>AAAI Conference on Artificial Intelligence</i> , 2026 |

- KDD '26 Blue Sky** 57. Yixiang Yao, Yuhang Yao, Xinyi Fan, **Jiechao Gao**, Jie Wang, Minjia Zhang, Srivatsan Ravi, Carlee Joe-Wong  
Trustworthy Agent Network: Trust in Agent Networks Must Be Baked In, Not Bolted On  
*ACM SIGKDD Conference on Knowledge Discovery and Data Mining, Blue Sky Track*, 2026
- WWW '26** 56. Sahil Tripathi, Gautam Siddharth Kashyap, Mehwish Nasim, Jian Yang, **Jiechao Gao**<sup>†</sup>, Usman Naseem  
They Said Memes Were Harmless-We Found the Ones That Hurt: Decoding Jokes, Symbols, and Cultural References  
*The ACM Web Conference*, 2026
- SIGIR '26** 55. Qi Sun, Yulin Xu, Zelin Wang, Xiaoyu Kang, Keyan Jin, **Jiechao Gao**<sup>†</sup>  
Distribution-aware Re-representations for Multi-Scenario Recommendations  
*International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2026
- ACL '26 Ind.** 54. Abdullah Mohammad, Sushant Kumar Ray, Pushkar Arora, Rafiq Ali, Ebad Shabbir, Gautam Siddharth Kashyap, **Jiechao Gao**<sup>†</sup>, Usman Naseem<sup>†</sup>  
Are Large Language Models Economically Viable for Industry Deployment?  
*Annual Meeting of the Association for Computational Linguistics, Industry Track*, 2026
- ACL '26 Ind.** 53. Soumedhik Bharati\*, Ebad Shabbir\*, **Jiechao Gao**<sup>†</sup>  
FROST: Factual Reasoning via Optimized Stochastic Trajectories in Large Language Models during Inference  
*Annual Meeting of the Association for Computational Linguistics, Industry Track*, 2026
- ACL Findings '26** 52. **Jiechao Gao**<sup>†</sup>, Rohan Kumar Yadav, Yuangang Li, Yuandong Pan, Jie Wang, Ying Liu, Michael Lepech  
LLM-Guided Semantic Bootstrapping for Interpretable Text Classification with Tsetlin Machines  
*Findings of the Association for Computational Linguistics: ACL 2026*
- EMNLP '26 Ind.** 51. Muran Yu, **Jiechao Gao**, Yuandong Pan, Barney Haoyun Miao, Andrew Carter Lesh, Kincho Law, Jie Wang, Michael Lepech  
EGT-KG: Evidence-Grounded Typed KG Retrieval for Practical Scientific QA with Small Language Models  
*Conference on Empirical Methods in Natural Language Processing, Industry Track*, 2026
- EACL '26 Ind.** 50. Sushant Kumar Ray, Gautam Siddharth Kashyap, Sahil Tripathi, Nipun Joshi, Vijay Govindarajan, Rafiq Ali, **Jiechao Gao**<sup>†</sup>, Usman Naseem  
Do Clinical Question Answering Systems Really Need Specialised Medical Fine Tuning?  
*Conference of the European Chapter of the Association for Computational Linguistics, Industry Track*, 2026
- EACL '26 Find.** 49. Gautam Siddharth Kashyap, Harsh Joshi, Niharika Jain, Ebad Shabbir, **Jiechao Gao**<sup>†</sup>, Nipun Joshi, Usman Naseem<sup>†</sup>  
Revealing the Truth with ConLLM for Detecting Multi-Modal Deepfakes  
*Conference of the European Chapter of the Association for Computational Linguistics*, 2026
- ICASSP '26** 48. Ebad Shabbir\*, Pushkar Arora\*, Rakshita Saksainaa, Tiange Xie, **Jiechao Gao**<sup>†</sup>  
AVATAR: Audio-Visual Adaptive Fusion via Trained Agent Reinforcement for Multimodal Deepfake Detection  
*IEEE International Conference on Acoustics, Speech and Signal Processing*, 2026
- ICASSP '26** 47. Yuheng Li\*, Taoyu Wu\*, **Jiechao Gao**<sup>†</sup>, Wei Zhu<sup>†</sup>, Wenwen Ouyang  
DemoReranker: Enhancing the In-Context Learning Capability of Multi-Modal Large Models via Demo Ranking  
*IEEE International Conference on Acoustics, Speech and Signal Processing*, 2026
- ICASSP '26** 46. Xinyuan Huang, **Jiechao Gao**<sup>†</sup>, Tiange Xie  
Differentiable Meta-Optimization for Federated Neural Architecture Search (MO-FedNAS)  
*IEEE International Conference on Acoustics, Speech and Signal Processing*, 2026
- ICASSP '26** 45. Taoyu Wu, Yuheng Li, **Jiechao Gao**<sup>†</sup>, Wei Zhu<sup>†</sup>, Ming Tan, Hui Yi Leong  
EPO: Enhanced Preference Optimization with Multi-Response Data for LLMs via Stochastic Softmax  
*IEEE International Conference on Acoustics, Speech and Signal Processing*, 2026
- ICML '25** 44. Shisong Tang\*, Chao Cui\*, Fan Li, **Jiechao Gao**, Hechang Chen  
Calibrating Video Watch-time Predictions with Credible Prototype Alignment  
*International Conference on Machine Learning*, 2025

- NeurIPS '25 43. Yidong Wu\*, Siyuan Chen\*, Binrui Wu, Fan Li, **Jiechao Gao**<sup>†</sup>  
Adaptive Gradient Masking for Balancing ID and MLLM-based Representations in Recommendation  
*Conference on Neural Information Processing Systems*, 2025
- KDD '25 42. Yulin Xu\*, Chao Cui\*, Shisong Tang\*, Fan Li, Bing Han, Huafeng Cao, **Jiechao Gao**<sup>†</sup>, Hechang Chen  
Leveraging Label Distributions as Anchors to Enhance Video Recommendation  
*ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2025
- KDD '25 41. Binrui Wu\*, Shisong Tang\*, Fan Li, Bing Han, Chang Meng, Jingyu Xiao, **Jiechao Gao**<sup>†</sup>  
Aligning and Balancing ID and Multimodal Representations for Recommendation  
*ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2025
- ACL Findings '25 40. Xunzhu Tang\*, **Jiechao Gao**\*, Jin Xu\*, Tiezhu Sun, Yewei Song, Saad Ezzini, Wendkūni C. Ouédraogo, Jacques Klein, Tegawendé F. Bissyandé  
SynFix: Dependency-Aware Program Repair via RelationGraph Analysis  
*Findings of the Association for Computational Linguistics: ACL 2025*
- EMNLP '25 Ind.  Best Paper Nom. 39. Mohammad Anas Azeez\*, Rafiq Ali\*, Ebad Shabbir, Zohaib Hasan Siddiqui, Gautam Siddharth Kashyap, **Jiechao Gao**<sup>†</sup>, Usman Naseem<sup>†</sup>  
Truth, Trust, and Trouble: Medical AI on the Edge  
*Conference on Empirical Methods in Natural Language Processing, Industry Track*, 2025
- EMNLP '25 Ind. 38. Zohaib Hasan Siddiqui\*, **Jiechao Gao**\*, Ebad Shabbir\*, Mohammad Anas Azeez, Rafiq Ali, Gautam Siddharth Kashyap, Usman Naseem<sup>†</sup>  
LLMs on a Budget? Say HOLA  
*Conference on Empirical Methods in Natural Language Processing, Industry Track*, 2025
- EMNLP '25 Ind. 37. Hui Yi Leong, Yuheng Li, Yuqing Wu, Wenwen Ouyang, Wei Zhu<sup>†</sup>, **Jiechao Gao**<sup>†</sup>, Wei Han  
AMAS: Adaptively Determining Communication Topology for LLM-based Multi-agent System  
*Conference on Empirical Methods in Natural Language Processing, Industry Track*, 2025
- EMNLP '25 Ind. 36. Yuheng Li\*, **Jiechao Gao**\*, Wei Han, Wenwen Ouyang, Wei Zhu<sup>†</sup>, Hui Yi Leong  
FT-MDT: Extracting Decision Trees from Medical Texts via a Novel Low-rank Adaptation Method  
*Conference on Empirical Methods in Natural Language Processing, Industry Track*, 2025
- IJCNLP-AAACL '25 35. Sahil Tripathi, MD Tabrez Nafis, Imran Hussain, **Jiechao Gao**<sup>†</sup>  
The Confidence Paradox: Can LLM Know When It Is Wrong?  
*International Joint Conference on Natural Language Processing and AAACL*, 2025
- ACM MM '25  Oral (top 4%) 34. Binrui Wu\*, Haochen Sui\*, Jiaye Lin\*, **Jiechao Gao**<sup>†</sup>, Ting Xu, Keyan Jin, Xuesong Zhang  
Prototype-Guided Representation Projection for Multi-Domain Multi-Task Recommendation  
*ACM International Conference on Multimedia*, 2025
- ACM MM '25 33. Songning Lai\*, Ninghui Feng\*, **Jiechao Gao**\*, Hao Wang, Haochen Sui, Xin Zou, Jiayu Yang, Wenshuo Chen, Lijie Hu, Hang Zhao, Xuming Hu, Yutao Yue  
From Guesswork to Guarantee: Towards Faithful Multimedia Web Forecasting with TimeSieve  
*ACM International Conference on Multimedia*, 2025
- CVPR '25 32. Yuzhi Huang\*, Chenxin Li<sup>†</sup>, Haitao Zhang, Zixu Lin, Yunlong Lin, Hengyu Liu, Wuyang Li, Xinyu Liu, **Jiechao Gao**, Yue Huang<sup>†</sup>, Xinghao Ding, Yixuan Yuan  
Track Any Anomalous Object: A Granular Video Anomaly Detection Pipeline  
*IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025
- ICCPS '25 31. **Jiechao Gao**\*, Mingyue Tang\*, Wenpeng Wang, Tushar Routh, Brad Campbell  
Atlas: Ensuring Accuracy for Privacy-Preserving Federated IoT Applications  
*ACM/IEEE International Conference on Cyber-Physical Systems*, 2025
- ICASSP '25 30. Gautam Kashyap, Zohaib Hasan Siddiqui, Mohammad Anas Azeez, Rafiq Ali, Shantanu Kumar, Navin Kamuni, **Jiechao Gao**<sup>†</sup>  
Fooling the Forgers: A Multi-Stage Framework for Audio Deepfake Detection  
*IEEE International Conference on Acoustics, Speech and Signal Processing*, 2025
- ICASSP '25 29. Gautam Kashyap, Harsh Joshi, Manaswi Kulahara, Rajkumar Dhakar, Atul Sajjanhar, **Jiechao Gao**<sup>†</sup>, Sarthak Jain, Shahab Sohail  
Can AI See What We Can't? Leveraging Deep Learning and Multi-Temporal Satellite Data to Revolutionize Crop Type Mapping and Yield Prediction  
*IEEE International Conference on Acoustics, Speech and Signal Processing*, 2025

- BigData '25** 28. **Jiechao Gao**, Rohan Kumar Yadav, Xinyuan Huang, Jie Wang  
Enhancing Interpretability in Self-Training with Tsetlin Machines for Mitigating Noisy Pseudo-Labels  
*IEEE International Conference on Big Data*, 2025
- BigData '25** 27. Xinyuan Huang, **Jiechao Gao**<sup>†</sup>, Jie Wang  
Federated Neural Architecture Search with Model-Agnostic Meta Learning  
*IEEE International Conference on Big Data*, 2025
- BIBM '25** 26. Xinyuan Huang, **Jiechao Gao**<sup>†</sup>  
Toward Fair and Efficient Neural Architecture Search in Heterogeneous Federated Learning  
*IEEE International Conference on Bioinformatics and Biomedicine*, 2025
- BIBM '25** 25. Xinyuan Huang, Hikaru Kurosawa, Jack Wunder, Sujit Patil, **Jiechao Gao**, Karthik Kuber, Jonathan C. Irish, Michael J. Daly  
U-KAN: Hybrid Spatial-Functional Deep Learning for Tumor Depth Estimation in Fluorescence-Guided Cancer Surgery  
*IEEE International Conference on Bioinformatics and Biomedicine*, 2025
- KSEM '25** 24. Huaibin Ye, Zuobin Ying, **Jiechao Gao**, Ximeng Liu, Jianping Cai  
FedBCE: Rethinking Clustered Federated Learning for Better Clustering Efficiency  
*International Conference on Knowledge Science, Engineering and Management*, 2025
- ASONAM '25** 23. Arpita Soni, Sahil Tripathi, Gautam Siddharth Kashyap, Manaswi Kulahara, Mohammad Anas Azeez, Zohaib Hasan Siddiqui, Nipun Joshi, **Jiechao Gao**<sup>†</sup>  
Can We Predict Your Next Move Without Breaking Your Privacy?  
*IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining*, 2025
- SEC '25** 22. Marshall Clyburn\*, Nabeel Nasir\*, Md Fazlay Rabbi Masum Billah, Victor Ariel Leal Sobral, **Jiechao Gao**, Fateme Nikseresht, Brad Campbell  
fReeLoaders: An IoT Ecosystem for Real-Time Deadline-Driven Task Scheduling using Reinforcement Learning  
*ACM/IEEE Symposium on Edge Computing*, 2025
- ACM MM '24** 21. Yiyong Xiao\*, Kai Shu\*, Haoyi Zhang\*, Baohua Yin, Wai Seng Cheang, Haoyang Wang, **Jiechao Gao**<sup>†</sup>  
EGGesture: Entropy-Guided Vector Quantized Variational AutoEncoder for Co-Speech Gesture Generation  
*ACM International Conference on Multimedia*, 2024
- BuildSys '24** 20. Tushar Routh, Nurani Saoda, Fateme Nikseresht, Md Fazlay Rabbi Masum Billah, **Jiechao Gao**,  
 **Best Paper Cand.**  
Viswajith Govinda Rajan, Bradford Campbell  
ScreenSense: Screen Activity Detection in Real-World Environments with Indoor Light Sensors  
*ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation*, 2024
- BigData '24** 19. Jiayu Wang, **Jiechao Gao**<sup>†</sup>  
Federated Learning with Knowledge Distillation to Mitigate Catastrophic Forgetting and Data Heterogeneity in IoV Systems  
*IEEE International Conference on Big Data*, 2024
- BIBM '24** 18. **Jiechao Gao**<sup>†</sup>, Yuangang Li  
FedMetaMed: Federated Meta-Learning for Personalized Medication in Distributed Healthcare Systems  
*IEEE International Conference on Bioinformatics and Biomedicine*, 2024
- BIBM '24** 17. Kai Shu\*, Yuzhuo Jia\*, Ziyang Zhang, **Jiechao Gao**<sup>†</sup>  
FODA-PG for Enhanced Medical Imaging Narrative Generation: Adaptive Differentiation of Normal and Abnormal Attributes  
*IEEE International Conference on Bioinformatics and Biomedicine*, 2024
- SciSec '24** 16. Tiange Xie, Feng Liu, **Jiechao Gao**, Yinghui Wang  
Integrating Consortium Blockchain and Attribute-Based Searchable Encryption for Automotive Threat Intelligence Sharing Model  
*International Conference on Science of Cyber Security*, 2024
- ICDM SSTDM '24** 15. **Jiechao Gao**<sup>†</sup>, Yuangang Li, Syeda Faiza Ahmed  
Fed-LDR: Federated Local Data-infused Graph Creation with Node-centric Model Refinement  
*IEEE ICDM International Workshop on Spatial and Spatiotemporal Data Mining*, 2024

- CHIL '23**  
 Oral (13.3%)  
 14. Mingyu Tang\*, **Jiechao Gao\***, Guimin Dong, Carl Yang, Bradford Campbell, Brendan Bowman, Jamie Marie Zoellner, Emaad Abdel-Rahman, Mehdi Boukhechba  
 SRDA: Mobile Sensing based Fluid Overload Detection for End Stage Kidney Disease Patients using Sensor Relation Dual Autoencoder  
*PMLR Conference on Health, Inference, and Learning*, 2023
- ICPP '23**  
 13. **Jiechao Gao**, Wenpeng Wang, Fateme Nikseresht, Viswajith Govinda Rajan, Bradford Campbell  
 PFDR: Personalized Federated Deep Reinforcement Learning for Residential Energy Management  
*International Conference on Parallel Processing*, 2023
- DCOSS '23**  
 12. Wenpeng Wang\*, Fateme Nikseresht\*, Viswajith Govinda Rajan, **Jiechao Gao**, Bradford Campbell  
 Enabling ubiquitous occupancy detection in smart buildings: A wifi ftm-based approach  
*International Conference on Distributed Computing in Smart Systems and the Internet of Things*, 2023
- PRCV '23**  
 11. Wei Cai, **Jiechao Gao**  
 SSCL: semi-supervised contrastive learning for industrial anomaly detection  
*Chinese Conference on Pattern Recognition and Computer Vision*, 2023
- NeurIPS D&B '22**  
 10. Xiao-Yang Liu\*, Ziyi Xia\*, Jingyang Rui, **Jiechao Gao**, Hongyang Yang, Ming Zhu, Christina Dan Wang<sup>†</sup>, Zhaoran Wang, Jian Guo<sup>†</sup>  
 FinRL-Meta: Market Environments and Benchmarks for Data-Driven Financial Reinforcement Learning  
*Conference on Neural Information Processing Systems, Datasets and Benchmarks Track*, 2022
- COMSNETS '22**  
 9. Priyan Malarvizhi Kumar, Bharat Rawal, **Jiechao Gao**  
 Blockchain-enabled privacy preserving of IoT data for sustainable smart cities using machine learning  
*International Conference on COMMunication Systems & NETWORKS*, 2022
- IPSN '21**  
 8. Md Fazlay Rabbi Masum Billah, Nurani Saoda, **Jiechao Gao**, Bradford Campbell  
 BLE can see: a reinforcement learning approach for RF-based indoor occupancy detection  
*ACM/IEEE International Conference on Information Processing in Sensor Networks*, 2021
- BuildSys '21**  
 7. Tong Wu\*, Murtadha Aldeer\*, Tahiya Chowdhury, Amber Haynes, Fateme Nikseresht, Mahsa Pahlavikhah Varnosfaderani, **Jiechao Gao**, Arsalan Heydarian, Brad Campbell, Jorge Ortiz  
 The smart building privacy challenge  
*ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation*, 2021
- BuildSys '21**  
 6. Wenpeng Wang, Zetian Liu, **Jiechao Gao**, Nurani Saoda, Bradford Campbell  
 UbiTrack: Enabling scalable & low-cost device localization with onboard wifi  
*ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation*, 2021
- ICPP '21**  
 5. Haoyu Wang, Haiying Shen, **Jiechao Gao**, Kevin Zheng, Xiaoying Li  
 Multi-agent reinforcement learning based distributed renewable energy matching for datacenters  
*International Conference on Parallel Processing*, 2021
- ICAIF '21**  
 4. Xiao-Yang Liu, Hongyang Yang, **Jiechao Gao**, Christina Wang  
 FinRL: Deep Reinforcement Learning Framework to Automate Trading in Quantitative Finance  
*ACM International Conference on AI in Finance*, 2021
- IPDPS '20**  
 3. **Jiechao Gao**, Haoyu Wang, Haiying Shen  
 Smartly handling renewable energy instability in supporting a cloud datacenter  
*IEEE International Parallel and Distributed Processing Symposium*, 2020
- ICCCN '20**  
 2. **Jiechao Gao**, Haoyu Wang, Haiying Shen  
 Machine learning based workload prediction in cloud computing  
*International Conference on Computer Communications and Networks*, 2020
- BigData '19**  
 1. **Jiechao Gao**, Haoyu Wang, Haiying Shen  
 Task failure prediction in cloud data centers using deep learning  
*IEEE International Conference on Big Data*, 2019
- Peer-reviewed Journal Papers**
- NPJ AI '26**  
 30. **Jiechao Gao\***, Yuangang Li\*, Jie Wang, Michael Lepech, Yue Zhao, Brad Campbell  
 H-FedSN: Personalized Sparse Networks for Efficient and Accurate Hierarchical Federated Learning for IoT Applications  
*NPJ Artificial Intelligence*, 2026
- IEEE TAI '26**  
 29. Gautam Siddharth Kashyap, Niharika Jain, Ebad Shabbir, Harsh Joshi, Usman Naseem, **Jiechao Gao**<sup>†</sup>  
 CLARITY: A Lightweight Multimodal Transformer for Harmful Content Detection  
*IEEE Transactions on Artificial Intelligence*, 2026

- AutoCon '26** 28. Yuandong Pan, Mudan Wang, **Jiechao Gao**, Jie Wang, Michael D. Lepech, Ioannis Brilakis  
LLM-enabled multi-agent framework for automated Scan-to-BIM and Scan-to-Graph reconstruction  
*Automation in Construction*, 2026
- AI for Eng. '25** 27. Keyi Wang, Nikolaus Holzer, Ziyi Xia, Yupeng Cao, **Jiechao Gao**, Anwar Walid, Kairong Xiao, Xiao-Yang Liu Yanglet  
FinRL Contests: Data-Driven Financial Reinforcement Learning Agents for Stock and Crypto Trading  
*Artificial Intelligence for Engineering*, 2025
- ACM TMIS '24** 26. Xiao-Yang Liu, Rongyi Zhu, Daochen Zha, **Jiechao Gao**, Shan Zhong, Meikang Qiu  
Differentially Private Low-Rank Adaptation of Large Language Model Using Federated Learning  
*ACM Transactions on Management Information Systems*, 2024
- EPSR '24** 25. Aamir Nawaz, Hongtao Wang, Huiting Yang, Hammad Armghan, **Jiechao Gao**<sup>†</sup>  
Risk-constrained probabilistic coordination in coupled transmission and distribution system  
*Electric Power Systems Research*, 2024
- Veh. Comm. '24** 24. Mehdi Gheisari, Aminreza Karamoozian, **Jiechao Gao**, Hemn Barzan Abdalla, Shuja Ansari, Riaz Ullah Khan, Zhaoxi Fang  
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- IEEE Access '24** 23. M. Mahad Malik, Abdullah Altamimi, Syed Ali Abbas Kazmi, Zafar Ali Khan, Muhammad Waleed Ansari, Kamran Mujahid, **Jiechao Gao**  
A Full-Fledged, Multi-Agent System Representing the Architecture of Smart Cities by Balancing Energy With Optimal Electricity Forecasting, Integrating Individual Comfort, and Extracting Financial Gains  
*IEEE Access*, 2024
- TPDS '23** 22. Haiying Shen, Haoyu Wang, **Jiechao Gao**, Rajkumar Buyya  
An Instability-Resilient Renewable Energy Allocation System for a Cloud Datacenter  
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- MLJ '23** 21. Xiao-Yang Liu, Ziyi Xia, Hongyang Yang, **Jiechao Gao**, Daochen Zha, Ming Zhu, Christina Dan Wang, Zhaoran Wang, Jian Guo  
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*Machine Learning Journal*, 2023
- ESWA '23** 20. Ahmad Almadhor, Rizwana Irfan, **Jiechao Gao**<sup>†</sup>, Nasir Saleem, Hafiz Tayyab Rauf, Seifedine Kadry  
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*Expert Systems with Applications*, 2023
- ASC '23** 19. Hafiz Tayyab Rauf, **Jiechao Gao**<sup>\*</sup>, Ahmad Almadhor, Ali Haider, Yu-Dong Zhang<sup>†</sup>, Fadi Al-Turjman  
Multi population-based chaotic differential evolution for multi-modal and multi-objective optimization problems  
*Applied Soft Computing*, 2023
- CAAI TIT '23** 18. Nasir Saleem, **Jiechao Gao**<sup>\*</sup>, Rizwana Irfan, Ahmad Almadhor, Hafiz Tayyab Rauf, Yudong Zhang, Seifedine Kadry  
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*CAAI Transactions on Intelligence Technology*, 2023
- JAIHC '23** 17. Mehdi Gheisari, Amir Javadpour, **Jiechao Gao**<sup>†</sup>, Aaqif Afzaal Abbasi, Quoc-Viet Pham, Yang Liu  
PPDMIT: A lightweight architecture for privacy-preserving data aggregation in the Internet of Things  
*Journal of Ambient Intelligence and Humanized Computing*, 2023
- JKE '23** 16. Morshadul Hasan, Thuhid Noor, **Jiechao Gao**, Muhammad Usman, Mohammad Zoynul Abedin  
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*Journal of the Knowledge Economy*, 2023
- TITS '22** 15. Gunasekaran Manogaran, **Jiechao Gao**<sup>\*</sup>, Tu N Nguyen  
Optimizing resource and service allocations for IoT-assisted intelligent transportation systems  
*IEEE Transactions on Intelligent Transportation Systems*, 2022
- TITS '22** 14. **Jiechao Gao**, Gunasekaran Manogaran, Tu N Nguyen, Seifedine Kadry, Ching-Hsien Hsu, Priyan Malarvizhi Kumar  
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*IEEE Transactions on Intelligent Transportation Systems*, 2022

- IoT-J '22** 13. Zeeshan Siddiqui, **Jiechao Gao**<sup>†</sup>, Muhammad Khurram Khan  
An Improved Lightweight PUF-PKI Digital Certificate Authentication Scheme for the Internet of Things  
*IEEE Internet of Things Journal*, 2022
- JBHI '22** 12. **Jiechao Gao**, Tu N Nguyen, Gunasekaran Manogaran, Ankit Chaudhary, Gai-Ge Wang  
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*IEEE Journal of Biomedical and Health Informatics*, 2022
- TNSE '22** 11. Gunasekaran Manogaran, Tu N Nguyen, **Jiechao Gao**<sup>†</sup>, Priyan Malarvizhi Kumar  
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*IEEE Transactions on Network Science and Engineering*, 2022
- ACM ToSN '22** 10. Guimin Dong, Mingyue Tang, Zhiyuan Wang, **Jiechao Gao**, Sikun Guo, Lihua Cai, Robert Gutierrez, Bradford Campbell, Laura E Barnes, Mehdi Boukhechba  
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*ACM Transactions on Sensor Networks*, 2022
- TGCN '22** 9. Tu N Nguyen, **Jiechao Gao**<sup>†</sup>, Gunasekaran Manogaran, R Dinesh Jackson Samuel, Mamoun Alazab  
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*IEEE Transactions on Green Communications and Networking*, 2022
- KBS '22** 8. **Jiechao Gao**<sup>\*†</sup>, Nasir Saleem\*, Muhammad Irfan Khattak, Hafiz Tayyab Rauf, Seifedine Kadry, Muhammad Shafi  
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- IVC '22** 7. Nasir Saleem, **Jiechao Gao**<sup>\*</sup>, Muhammad Irfan, Elena Verdu, Javier Parra Fuente  
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*Image and Vision Computing*, 2022
- CEE '22** 6. Muhammad Irfan Khattak, Nasir Saleem, **Jiechao Gao**, Elena Verdu, Javier Parra Fuente  
Regularized sparse features for noisy speech enhancement using deep neural networks  
*Computers and Electrical Engineering*, 2022
- Agriculture '22** 5. Yuanzhi Pan, Hua Jin, **Jiechao Gao**<sup>†</sup>, Hafiz Tayyab Rauf  
Identification of Buffalo Breeds Using Self-Activated-Based Improved Convolutional Neural Networks  
*Agriculture*, 2022
- Neurocomp. '22** 4. Adeel Ahmed, Khalid Saleem, Osman Khalid, **Jiechao Gao**<sup>†</sup>, Umer Rashid  
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*Neurocomputing*, 2022
- JBHI '21** 3. Priyan Malarvizhi Kumar, Choong Seon Hong, Fatemeh Afghah, Gunasekaran Manogaran, Keping Yu, Qiaozhi Hua, **Jiechao Gao**  
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*IEEE Journal of Biomedical and Health Informatics*, 2021
- FGCS '21** 2. Mehdi Gheisari, Hamid Esmaeili Najafabadi, Jafar A Alzubi, **Jiechao Gao**<sup>†</sup>, Guojun Wang, Aaqif Afzaal Abbasi, Aniello Castiglione  
OBPP: An ontology-based framework for privacy-preserving in IoT-based smart city  
*Future Generation Computer Systems*, 2021
- TSC '20** 1. **Jiechao Gao**, Haoyu Wang, Haiying Shen  
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- Peer-reviewed Workshop, Poster & Demo Papers**
- ICWSM W '26** 11. Daniel Koo, **Jiechao Gao**, Yuandong Pan, Jie Wang, Michael D. Lepech  
Perceiving Creativity in the Age of AI: How Labels, Beliefs, and Familiarity Shape Evaluations of AI-Generated and Human-Created Art  
*Workshop on Social Influence in the Era of LLMs (SocialLLM) at ICWSM*, 2026
- ICCPS '25 Poster** 10. Tushar Routh, **Jiechao Gao**, Bradford Campbell  
Data Efficient PV based Indoor Event Detection  
*ACM/IEEE International Conference on Cyber-Physical Systems, Poster Track*, 2025

- ACL '25 SRW 9. Ellen Yi-Ge, **Jiechao Gao**, Wei Han, Wei Zhu<sup>†</sup>  
 DRUM: Learning Demonstration Retriever for Large Multi-modal Models  
*Annual Meeting of the Association for Computational Linguistics Student Research Workshop*, 2025
- ACL '25 SRW 8. Juyuan Zhang, **Jiechao Gao**, Wenwen Ouyang, Wei Zhu<sup>†</sup>, Hui Yi Leong  
 Time-Llama: Adapting Large Language Models for Time Series Modeling via Dynamic Low-rank Adaptation  
*Annual Meeting of the Association for Computational Linguistics Student Research Workshop*, 2025
- NeurIPS W '25 7. Chunlin Feng, Lijian Huang, Keyi Wang, Yupeng Cao, Ming Zhu, **Jiechao Gao**, Xiao-Yang Liu  
 Standard Market Environments for Financial Reinforcement Learning  
*NeurIPS 2025 Workshop on Generative AI in Finance*, 2025
- AAAI Bridge '23 6. Berend Jelmer Dirk Gort\*, Xiao-Yang Liu\*, Xinghang Sun, **Jiechao Gao**, Shuaiyu Chen, Christina Dan Wang.  
 Deep Reinforcement Learning for Cryptocurrency Trading: Practical Approach to Address Backtest Overfitting  
*AAAI Bridge Program on AI for Financial Services*, 2023
- SenSys '22 Poster 5. **Jiechao Gao**\*, Mingyue Tang\*, Tianhao Wang, Bradford Campbell  
 Pfed-ldp: A personalized federated local differential privacy framework for iot sensing data  
*ACM Conference on Embedded Networked Sensor Systems, Poster Track*, 2022
- IPSN '22 Poster 4. **Jiechao Gao**, Wenpeng Wang, Bradford Campbell  
 Residential Energy Management System Using Personalized Federated Deep Reinforcement Learning  
*ACM/IEEE International Conference on Information Processing in Sensor Networks, Poster Track*, 2022
- NeurIPS '21 W 3. Xiao-Yang Liu, Jingyang Rui, **Jiechao Gao**, Liuqing Yang, Hongyang Yang, Zhaoran Wang, Christina Dan Wang, Jian Guo  
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*Conference on Neural Information Processing Systems*, 2021
- GLOBECOM W '21 2. M Poongodi, Mounir Hamdi, **Jiechao Gao**, Hafiz Tayyab Rauf  
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*IEEE Global Communications Conference Workshops*, 2021
- SenSys W '21 1. **Jiechao Gao**, Wenpeng Wang, Zetian Liu, Md Fazlay Rabbi Masum Billah, Bradford Campbell  
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*ACM Conference on Embedded Networked Sensor Systems, AIChallengeIoT Workshop*, 2021
- Peer-reviewed Book Chapters**
- Book '22 3. **Jiechao Gao**, Tiange Xie  
 DNA computing in cryptography  
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- Book '20 2. Hongzhuo Qi, Yi Qin, **Jiechao Gao**  
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- Book '19 1. Jinglan Wu, Dongliang Lin, **Jiechao Gao**, Ye Xu  
 Introduction of Bigdata and Artificial Intelligence Applications  
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- Preprints**
- arXiv '26 14. When Properties Are Not Separated: Do LLMs Fail at Judging?
- arXiv '26 13. Clinically Grounded Curriculum Learning for Radiology Report Generation
- arXiv '26 12. Fine-Grained Temporal Data Augmentation for Human Activity Recognition
- arXiv '26 11. GRM: Graph Reward Modeling for Multimodal Retrieval-Augmented Generation
- arXiv '26 10. Matched-Corruption Decoding: Tracing Support in Visual Reasoning
- arXiv '26 9. Show Your Evidence: Evaluating Grounded Instructional Video with Video-LLMs
- arXiv '26 8. Harbor Adapters and Harbor-Index: Infrastructure and a Curated Meta-Dataset for Large-Scale Agentic Evaluation

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|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| arXiv '26                   | 7. PrivateHub: Contrastive Diffusion Model for Private Sensor-Intensive Environment Data Generation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| arXiv '26                   | 6. The Orchestration Gap: Why Process Automation Stalls in Operationally Complex Industries                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| arXiv '26                   | 5. Clusters are All You Need: Pre-Training the Tsetlin Machine with Semantic Clusters from Language Models for Interpretability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| arXiv '25                   | 4. ChildGuard: A Specialized Dataset for Combatting Child-Targeted Hate Speech                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| arXiv '25                   | 3. Can Argus Judge Them All? Comparing VLMs Across Domains                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Preprints.org '24           | 2. Artificial Intelligence-Aided Digital Twin Design: A Systematic Review and Future Directions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| arXiv '23                   | 1. From Simulations to Reality: Enhancing Multi-Robot Exploration for Urban Search and Rescue                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| GRANT & PROPOSAL EXPERIENCE | <p>Contributed to funded research and proposal preparation as a graduate researcher / postdoc (not PI).</p> <p><b>NSF: 2217071</b> Jul. 2022 - Sep. 2027<br/> Collaborative Research: <b>PPoSS: LARGE: Co-designing Hardware, Software, and Algorithms to Enable Extreme-Scale Machine Learning Systems</b><br/> PI/co-PI(s): Aidong Zhang, Matthew Dwyer, Kevin Skadron, Bradford Campbell, Ashish Venkat</p> <p><b>USDOT</b> <i>under submission</i><br/> University Transportation Centers Program - Tier 1 Center<br/> CIFE</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| STUDENT MENTORSHIP          | <p>20+ students have worked with me on research projects, many of whom moved on to PhD programs. Venues in parentheses are papers they first-authored or co-authored during our collaboration.</p> <ul style="list-style-type: none"> <li>• Muran Yu: Stanford PhD (EMNLP'26 Ind.)</li> <li>• Yulin Xu: UCI CS MS → USC CS PhD (KDD'25, ICML'26, SIGIR'26)</li> <li>• Yuangang Li: USC CS MS → UCI CS PhD (AAAI'26, Findings of ACL'26, NPJ AI'26, BIBM'24, ICDM'24 W)</li> <li>• Tiange Xie: Columbia U MS → IIE@CAS PhD (SciSec'24, ICASSP'26 ×2, book chapter)</li> <li>• Mingyue Tang: USC CS MS → UVA SE PhD → UIUC CS PhD (ICCPs'25, SenSys'22 Poster)</li> <li>• Gautam Siddharth Kashyap: Jamia Hamdard → Macquarie University CS PhD (EACL'26, ICASSP'26 ×2, ACL'26, WWW'26, EMNLP'25)</li> <li>• Berend Gort: Columbia U MLE → Universitat Politècnica de Catalunya CS PhD</li> <li>• Yuanbo Li: Zhengzhou University CS PhD</li> <li>• Kai Shu: USC CS MS (BIBM'24, ACM MM'24)</li> <li>• Xinyuan Huang: U of Toronto CS MS → U of Toronto CS PhD (ICASSP'26, BigData'25, BIBM'25)</li> <li>• Binrui Wu: Fudan University (KDD'25, ACM MM'25 Oral, NeurIPS'25, ICML'26)</li> <li>• Siyuan Chen: University of Bristol (ICLR'26, NeurIPS'25)</li> <li>• Haoran Ma: Harvard University (ICML'26)</li> <li>• Linxiao Li: University of Sydney (ICML'26)</li> <li>• Haochen Sui: University of Michigan (ACM MM'25 ×2)</li> <li>• Qi Sun: University of Melbourne (SIGIR'26)</li> <li>• Yidong Wu: Imperial College London (NeurIPS'25)</li> <li>• Ebad Shabbir: DSEU-Okhla → MBZUAI (ACL'26 ×2, EACL'26, EMNLP'25 ×2, ICASSP'26)</li> <li>• Rafiq Ali: DSEU-Okhla → MBZUAI (ACL'26, EACL'26, EMNLP'25 ×2, ICASSP'26)</li> <li>• Harsh Joshi: Bharati Vidyapeeth's College of Engineering → UC San Diego MS (EACL'26, ICASSP'26)</li> <li>• Daniel Koo: The Nueva School (high school) (ICWSM'26 W)</li> </ul> |

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|--------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------|-----------|
| TALKS AND LECTURES | The Chinese University of Hong Kong, Shenzhen | <i>Translational AI Systems: What Has to Be True Before Anyone Can Rely on What a Model Produces</i>        | Apr. 2026 |
|                    | EMNLP 2025 Industry Track, Suzhou             | <i>Truth, Trust, and Trouble: Medical AI on the Edge</i> (Best Paper Nomination)                            | Nov. 2025 |
|                    | ACM Multimedia 2025, Dublin                   | <i>Prototype-Guided Representation Projection for Multi-Domain Multi-Task Recommendation</i> (Oral, top 4%) | Oct. 2025 |
|                    | Stanford University                           | <i>Private and Accurate IoT Applications</i>                                                                | May. 2025 |
|                    | ICCPs 2025, Irvine                            | <i>Atlas: Ensuring Accuracy for Privacy-Preserving Federated IoT Applications</i>                           | May. 2025 |
|                    | Rutgers Efficient AI Seminar                  | <i>Federated Learning Enabled IoT Systems</i>                                                               | May. 2024 |
|                    | ICPP 2023, Salt Lake City                     | <i>PFDRRL: Personalized Federated Deep Reinforcement Learning for Residential Energy Management</i>         | Aug. 2023 |
|                    | CHIL 2023, New York                           | <i>SRDA: Mobile Sensing based Fluid Overload Detection for End Stage Kidney Disease Patients</i> (Oral)     | Jun. 2023 |
|                    | Commonwealth Cyber Initiative Symposium       | <i>PFed-LDP: A Personalized Federated Local Differential Privacy Framework for IoT Sensing Data</i>         | Apr. 2023 |
|                    | Beijing Institute of Technology               | <i>Towards Accurate and Private Cloud-IoT Applications</i>                                                  | Feb. 2023 |
|                    | Jilin University                              | <i>Towards Accurate and Private Cloud-IoT Applications</i>                                                  | Feb. 2023 |
|                    | Beijing University of Technology              | <i>Towards Accurate and Private Cloud-IoT Applications</i>                                                  | Feb. 2023 |
|                    | UVA ML Seminar                                | <i>Federated Learning with Privacy</i>                                                                      | Aug. 2022 |
|                    | UVA Link Lab                                  | <i>Towards Accurate and Private IoT Applications</i>                                                        | Feb. 2022 |
|                    | IPDPS 2020 (virtual)                          | <i>Smartly Handling Renewable Energy Instability in Supporting a Cloud Datacenter</i>                       | May. 2020 |
|                    | Frees Fund                                    | <i>How Neural Networks and Deep Learning Work in the Financial Industry</i>                                 | Aug. 2018 |

|                     |                                                                              |                     |
|---------------------|------------------------------------------------------------------------------|---------------------|
| TEACHING EXPERIENCE | <b>Stanford University</b>                                                   | Stanford, CA        |
|                     | <b>Guest Lecture</b>                                                         | Winter, 2026        |
|                     | <i>Stanford University CEE 221</i>                                           |                     |
|                     | <b>University of Virginia</b>                                                | Charlottesville, VA |
|                     | <b>Teaching Assistant</b>                                                    | Fall 2020           |
|                     | <i>Computer Networks</i> (Prof. Brad Campbell)                               |                     |
|                     | <b>Teaching Assistant</b> (helping on course topics)                         | Spring 2020         |
|                     | <i>Advanced Computer Networks</i> (Prof. Yixin Sun)                          |                     |
|                     | <b>Teaching Assistant</b>                                                    | Fall 2019           |
|                     | <i>Principles of Modeling for Cyber-Physical Systems</i> (Prof. Madhur Behl) |                     |
|                     | <b>Columbia University</b>                                                   | New York City, NY   |
|                     | <b>Teaching Assistant</b>                                                    | Fall 2017           |
|                     | <i>Reinforcement Learning</i> (Prof. Chong Li)                               |                     |

|          |                                                             |                  |
|----------|-------------------------------------------------------------|------------------|
| SERVICES | <b>Conference/Workshop/Competition Organizing Committee</b> |                  |
|          | • ASONAM Multidisciplinary Track Co-Chair                   | 2026             |
|          | • FinRL Contest ICAIF                                       | 2023, 2024, 2025 |
|          | • UbiComp/ISWC Workshop on Design with Autistic Children    | 2024             |
|          | • SecureComm Workshop on S/P-IoT                            | 2022             |

#### Area Chair and/or (Senior) Program Committee/Reviewer for Conferences and Workshops

- |                                        |                              |
|----------------------------------------|------------------------------|
| • IJCAI 2025 (SPC)                     | • SIGIR 2026                 |
| • AAAI-AIA 2025 (SPC)                  | • BMVC 2026                  |
| • IJCNN 2025, 2026 (AC)                | • ICML 2025, 2026            |
| • EACL Industry Track 2026 (AC)        | • ACL ARR 2026               |
| • IEEE IDS 2025, 2026 (Track Chair)    | • KDD 2023, 2024, 2025, 2026 |
| • NeurIPS 2022, 2023, 2024, 2025, 2026 | • AAAI 2024, 2025, 2026      |
| • EMNLP Industry 2026                  | • SDM 2024                   |

- ML4H 2023, 2024
- UbiComp/IMWUT 2025
- ICLR 2025, 2026
- MLHC 2025
- ICASSP 2025, 2026 (AC)
- CVPR 2025, 2026
- ICCV 2025
- ICCPS 2025
- ACM MM 2025, 2026
- DSNE 2023–2026
- SecureComm 2022
- 2nd QTNML@NeurIPS 2022
- DCAA@AAAI 2023
- MILET@SIGKDD 2023
- AI4TS@IJCAI 2023
- Sensys Workshop on IoT Datasets 2024
- AI4TS@SDM 2024

### Journal Reviewer

- Social Network Analysis and Mining (Associate Editor)
- IEEE Transactions on Systems, Man and Cybernetics: Systems
- Journal of Machine Learning Research (JMLR)
- IEEE Internet of Things Journal (IoT-J)
- IEEE Transactions on Intelligent Transportation Systems (TITS)
- IEEE Journal of Biomedical and Health Informatics (JBHI)
- IEEE Transactions on Cloud Computing (TCC)
- ACM Transactions on Sensor Networks (ToSN)
- Journal of Industrial Information Integration
- ACM Transactions on Asian and Low-Resource Language Information Processing (TALLIP)
- Financial Innovation
- ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)
- Expert Systems with Applications (ESWA)
- Cluster Computing
- IEEE Sensors Journal
- IEEE Access
- Soft Computing
- Journal of Ambient Intelligence and Humanized Computing
- Multimedia Tools and Applications

### REFERENCES

#### Prof. Bradford Campbell

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#### Prof. Michael Lepech

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 Stanford University  
 Reference email: [mlepech@stanford.edu](mailto:mlepech@stanford.edu)

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 Department of Civil and Environmental Engineering  
 Stanford University  
 Reference email: [jiewang@stanford.edu](mailto:jiewang@stanford.edu)